Responsible Machine Learning

Lecture 5: Machine Learning Model Debugging

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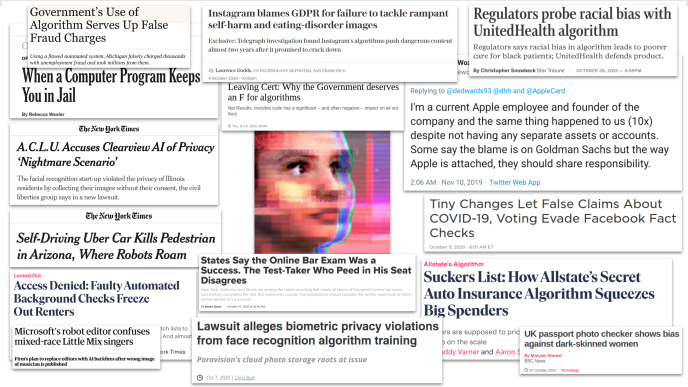
Acknowledgements

What is Model Debugging?

- Model debugging is an emergent discipline focused on discovering and remediating errors in the internal mechanisms and outputs of machine learning models.*
- Model debugging attempts to test machine learning models like software (because the models are software).
- Model debugging is similar to model validation and regression diagnostics, but for machine learning models.
- Model debugging **promotes trust directly** and **enhances interpretability as a side-effect**.

*See https://debug-ml-iclr2019.github.io/ for numerous examples of model debugging approaches.

Why Debug?

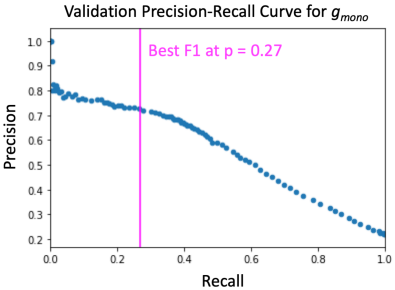


AI incidents: The AI Incident Database contains over 2,000 incident reports.†

†See https://incidentdatabase.ai/ to access the database.

Why Debug?

- Constrained, monotonic GBM probability of default (PD) classifier, g_{mono}.
- Grid search over hundreds of models.
- Best model selected by validation-based early stopping.
- Seemingly well-regularized (row and column sampling, explicit specification of L1 and L2 penalties).
- No evidence of over- or under-fitting.
- Better validation logloss than benchmark GLM.
- Decision threshold selected by maximization of F1 statistic.
- BUT **traditional assessment can be inadequate!**

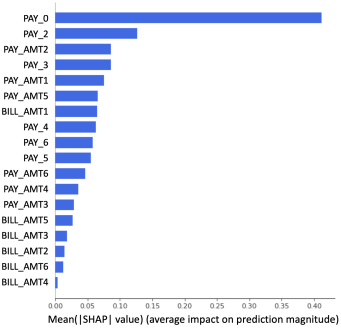


Validation Confusion Matrix at Threshold:

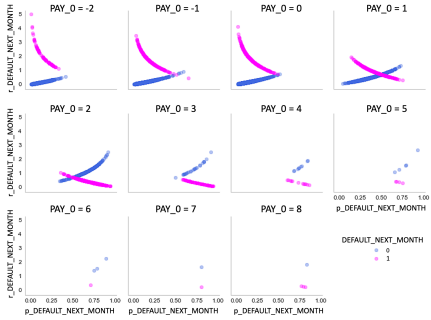
	Actual: 1	Actual: 0
Predicted: 1	1159	827
Predicted: 0	1064	6004

Why Debug?

Machine learning models can be **unnecessary**.



`gmono` PD classifier over-emphasizes the most important feature, a customer's most recent repayment status, PAY_0.



`gmono` also struggles to predict default for favorable statuses, $-2 \leq$ PAY_0 < 2, and often cannot predict on-time payment when recent payments are late, PAY_0 ≥ 2.

Why Debug?

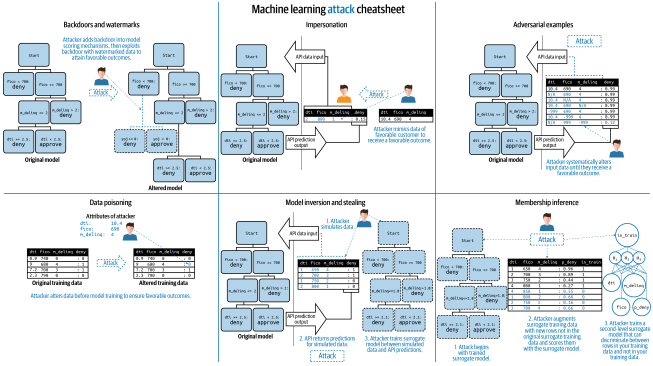
Machine learning models can perpetuate **sociological biases** [1].

	Adverse Impact Disparity	Accuracy Disparity	True Positive Rate Disparity	Precision Disparity	Specificity Disparity
single	0.885	1.029	0.988	1.008	1.025
divorced	1.014	0.932	0.809	0.806	0.958
other	0.262	1.123	0.62	1.854	1.169

Group disparity metrics are out-of-range for g_{mono} across different marital statuses.

Why Debug?

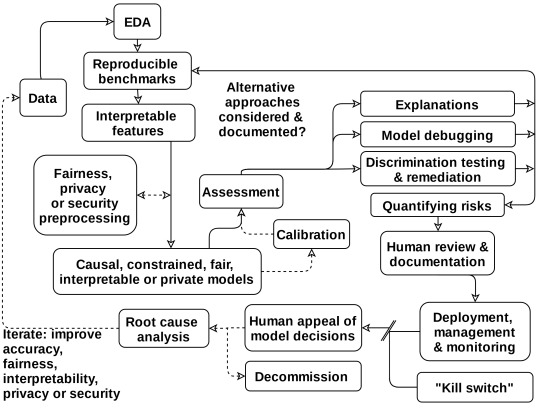
Machine learning models can have **security vulnerabilities** [2], [6], [7].‡



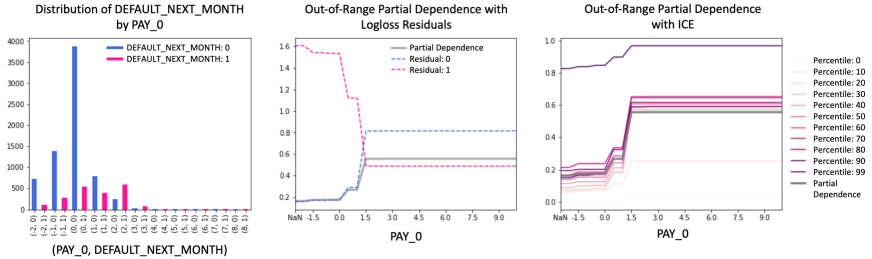
‡See https://bit.ly/3jyYtzi for full size image.

How to Debug Models?

As part of a holistic, low-risk approach to machine learning [4].

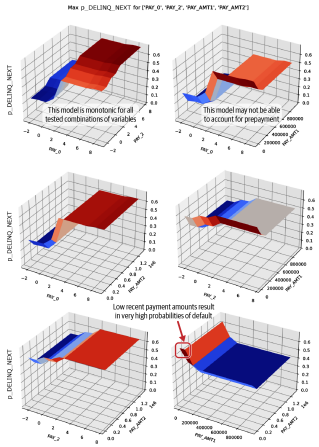


Sensitivity Analysis: Partial Dependence and ICE



- Training data is very sparse for PAY_0 > 2.
- ICE curves indicate that partial dependence is likely trustworthy and empirically confirm monotonicity, but also expose adversarial attack vulnerabilities.
- Partial dependence and ICE indicate g_{mono} likely learned very little for PAY_0 ≥ 2.
- PAY_0 $=$ `missing` gives lowest probability of default.

Sensitivity Analysis: Search for Adversarial Examples



Adversary search confirms multiple avenues of attack and exposes a potential flaw in g_{mono} inductive logic: default is predicted for customer's who make payments above their credit limit. (Try heuristics, evolutionary learning or packages like cleverhans to generate adversarial examples.)

Sensitivity Analysis: Robustness to Drift

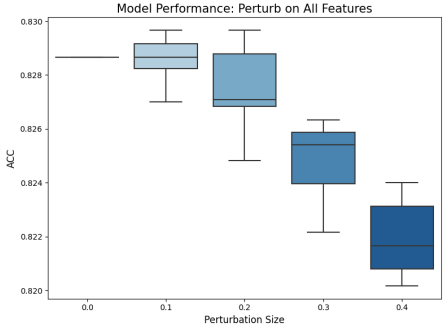
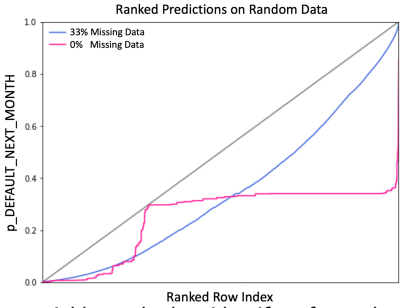


Figure: g_{mono} accuracy under feature perturbation.

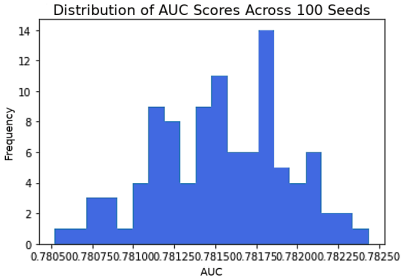
- Models must be robust to data drift once deployed.
- Simulation, perturbation, and statistics like population stability index (PSI), t, and Kolmogorov-Smirnov (K-S) can help assess robustness.
- Drift can also be measured on a feature-by-feature basis across data partitions.
- Likely due to monotonicity contraints g_{mono} holds up well to moderate data perturbation.

Sensitivity Analysis: Random Attacks



- In general, random attacks are a viable method to identify software bugs in machine learning pipelines. **(Start here if you don't know where to start.)**
- Random data can apparently elicit all probabilities $\in [0, 1]$ from g_{mono}.
- Around the decision threshold, lower probabilities can be attained simply by injecting missing values, yet another vulnerability to adversarial attack.
- Chaos testing is a broader approach that can also elicit unexpected approaches from machine learning systems.

Sensitivity Analysis: Underspecification



- Without domain-informed constraints ML models suffer from *underspecification* [3].
- Explicit tests for underspecification involve assessing model performance stability across perturbed computational hyperparameters: seeds, threads, number of GPUs, etc.
- Likely due to monotonicity constraints, g_{mono} performance appears stable across random seeds.

Residual Analysis: Segmented Error Analysis

Error Metrics for PAY_0

PAY_0	Prevalence	Accuracy	True Positive Rate	Precision	Specificity	Negative Predicted Value	False Positive Rate	False Discovery Rate	False Negative Rate	False Omissions Rate
-2	0.124	0.864	0.099	0.333	0.972	0.884	0.028	0.667	0.901	0.116
-1	0.168	0.816	0.206	0.406	0.939	0.854	0.061	0.594	0.794	0.146
0	0.121	0.867	0.107	0.341	0.972	0.888	0.028	0.659	0.893	0.112
1	0.325	0.491	0.903	0.381	0.292	0.862	0.708	0.619	0.097	0.138
2	0.709	0.709	1	0.709	0	0.5	1	0.291	0	0.5
3	0.748	0.748	1	0.748	0	0.5	1	0.252	0	0.5
4	0.571	0.571	1	0.571	0	0.5	1	0.429	0	0.5
5	0.444	0.444	1	0.444	0	0.5	1	0.556	0	0.5
6	0.25	0.25	1	0.25	0	0.5	1	0.75	0	0.5
7	0.5	0.5	1	0.5	0	0.5	1	0.5	0	0.5
8	0.75	0.75	1	0.75	0	0.5	1	0.25	0	0.5

Error Metrics for SEX

SEX										
Male	0.235	0.782	0.626	0.531	0.83	0.879	0.17	0.469	0.374	0.121
Female	0.209	0.797	0.552	0.514	0.862	0.879	0.138	0.486	0.448	0.121

- Notable change in accuracy and error characteristics for PAY_0 $\geq$ 2.
- For SEX, accuracy and error characteristics vary little across individuals represented in the training data. Bias mitigation should be confirmed by more involved bias testing.
- Overfitting, stability and other characteristics should also be analyzed by segment.
- Varying performance across segments can be an indication of underspecification.

Residual Analysis: Plotting Residuals

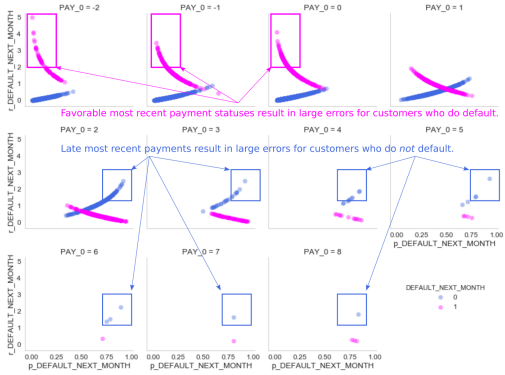
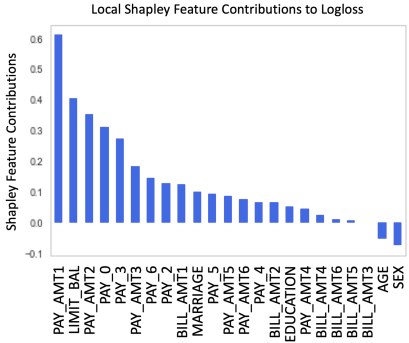


Figure: Residuals plotted by PAY_0 reveal a serious problem with g_{mono}.

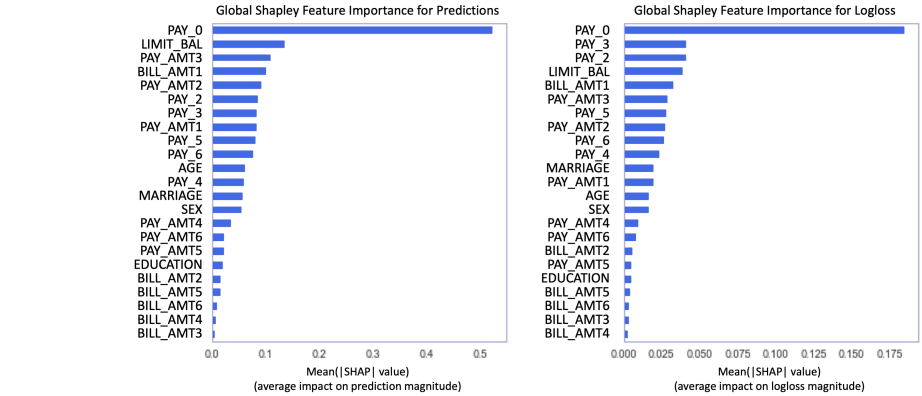
- Plotting residuals is a battle-tested model debugging technique.
- Residuals can be plotted using many approaches:
  - Overall, by feature (at left) or by segment
  - Traditional ($\hat{y}^{(i)} - y^{(i)}$)
  - Deviance or loss residuals (at left)
- Residuals can reveal serious issues and the underlying problems behind them.

Residual Analysis: Local Contributions to Logloss



Exact, local feature contributions to logloss can be calculated, enabling ranking of features contributing to logloss residuals for **each prediction**. Shapley contributions to XGBoost logloss can be calculated using the shap package. This is a **time-consuming** calculation.

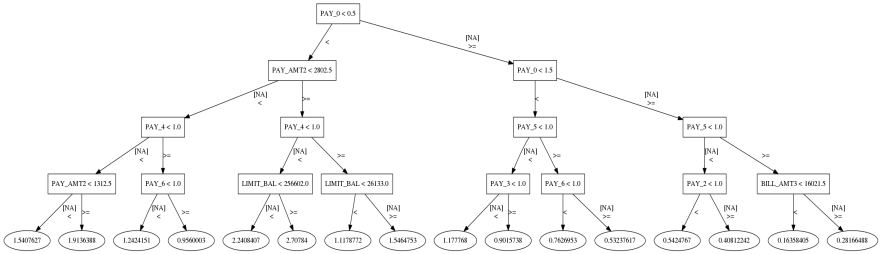
Residual Analysis: Non-Robust Features



Globally important features PAY_3 and PAY_2 are more important, on average, to the loss than to the predictions.

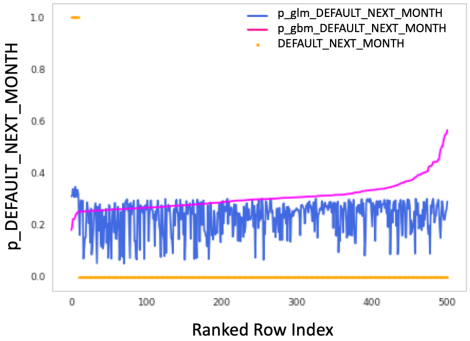
Residual Analysis: Modeling Residuals

Decision tree model of g_{mono} DEFAULT_NEXT_MONTH $= 1$ logloss residuals with 3-fold CV MSE $= 0.0070$ and $R^2 = 0.8871$.



This tree encodes rules describing when g_{mono} is probably wrong.

Benchmark Models: Compare to Linear Models



For a range of probabilities $\in (\sim 0.2, \sim 0.6)$, g_{mono} displays exactly incorrect prediction behavior as compared to a benchmark GLM.

Remediation: g_{mono}

- **Over-emphasis of PAY_0**:
  - Engineer features for payment trends or stability.
  - Strong regularization or missing value injection during training or inference.
- **Sparsity of PAY_0 > 2 training data**: Increase observation weights.
- **Payments $\geq$ credit limit**: Inference-time model assertion [5].
- **Disparate impact**: Adversarial de-biasing [8] or model selection by minimal disparate impact.
- **Security vulnerabilities**: API throttling, authentication, real-time model monitoring.
- **Large logloss importance**: Evaluate dropping non-robust features.
- **Poor accuracy vs. benchmark GLM**: Blend g_{mono} and GLM for probabilities $\in (\sim 0.2, \sim 0.6)$.
- **Miscellaneous strategies**:
  - Local feature importance and decision tree rules can indicate additional inference-time model assertions, e.g., alternate treatment for locally non-robust features in known high-residual ranges of the learned response function.
  - Incorporate local feature contributions to logloss into training or inference processes.

Remediation: General Strategies

Technical:

- Calibration to past data
- Data augmentation
- Discrimination remediation
- Experimental design
- Interpretable models
- Model or model artifact editing
- Model assertions
- Model monitoring
- Monotonicity and interaction constraints
- Strong regularization or missing value injection during training or inference

Process:

- Appeal and override
- Bug bounties
- Demographic and professional diversity
- Domain expertise
- Incident response plans
- Model risk management
  - Effective challenge and human review
- Software quality assurance
- Red-teaming

Acknowledgments

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References

This presentation:
https://www.github.com/jphall663/jsm_2019

Code examples for this presentation:
https://www.github.com/jphall663/interpretable_machine_learning_with_python
https://www.github.com/jphall663/responsible_xai

References

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[4] Patrick Hall et al. "A Responsible Machine Learning Workflow with Focus on Interpretable Models, Post-hoc Explanation, and Discrimination Testing." In: *Information* 11 (3 2020). URL: https://www.mdpi.com/2078-2489/11/3/137.

[5] Daniel Kang et al. *Debugging Machine Learning Models via Model Assertions*. URL: https://debug-ml-iclr2019.github.io/cameraready/DebugML-19_paper_27.pdf.

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